

Nimbus Nine — SD1 Training Pilot

Workbook 3

Merge, Append, and Validate

Difficulty	4 / 10 (rewarding)
Estimated time	60 minutes

Files for this workbook

Start file: 02_workbook_data/workbook3_start.xlsx

Complete file: 02_workbook_data/workbook3_complete.xlsx

Source files required:

- 02_workbook_data/workbook3_start.xlsx

FROM THE FLIGHT DECK

Captain Vega Sparrowsong says:

“Joining data is when it all comes together. We have got this. 🍷”

YOU ARE DEVELOPING

Skills: S4 (combine data sources), S5 (assess quality), S6 (use software tools)

Knowledge: K7 (data sources), K8 (data formats), K11 (data cleansing techniques)

Behaviours: B2 (works methodically), B3 (works collaboratively), B4 (responds positively to feedback)

The mission

MISSION BRIEF

We have three clean queries from Workbook 2.

Now we will join them together using Power Query merges, try a small append, and check the result.

When we are done, our final clean output will live in
02_workbook_data/sd1_complete.xlsx — ready for Captain Vega.

Two new words

A merge brings columns from one table into another. Think of it as adding wider rows.

An append stacks rows underneath rows from a similar table. Think of it as adding longer columns.

WHY THIS MATTERS

Most real reports come from joining several tables.

Power Query lets you do this without writing any code or VLOOKUPs.

It also quietly preserves your join logic so anyone can re-run it tomorrow.

Mission 1 — Understand merge versus append

Quick mental model:

- Merge = look up extra columns from a related table. Wider table.
- Append = stack new rows on top. Longer table.
- We will mostly merge today, with one small append at the end.

Join types in plain English

- Left Join — keep ALL rows from the left table. Extra columns from the right show as blank when there is no match.
- Inner Join — keep only rows that match in BOTH tables. Tighter, but you lose unmatched rows.
- Left Join is usually safest when the left table is the one you care about.
- Right Joins and Full Outer Joins exist too. You will meet them later.

Mission 2 — Open 02_workbook_data/workbook3_start.xlsx and reopen Power Query safely

Power Query should still be available from your previous work. We will make sure.

1. Open 02_workbook_data/workbook3_start.xlsx.
2. If Power Query Editor is already open, continue from the open window.
3. If Power Query Editor is not open: click Data, then Queries & Connections, then double-click the query you need (Deliveries_Clean).
4. If a query is missing: click Data, then Get Data, import the source again, and click Transform Data.

PAUSE AND CHECK

You should be able to see Deliveries_Clean, Couriers_Clean, and Routes_Clean in the Queries & Connections panel.

If any are missing, follow the recovery steps above.

Mission 3 — Merge Deliveries_Clean with Routes_Clean

We will add the route_name and difficulty into our deliveries view.

1. On the Home tab, click Merge Queries.
2. Choose Merge Queries (not Merge Queries as New) for now.
3. In the upper preview, click the route_code column to select it as the join key.
4. In the lower drop-down, choose Routes_Clean.
5. In the lower preview, click the route_code column to select it as the matching key.
6. Under Join Kind, choose Left Outer (all from first, matching from second).
7. At the bottom you should see a hint like '49 of 50 rows match'.
8. Click OK.
9. A new column appears at the right called Routes_Clean. It contains a Table value in each row.

COMMON TRAP

If the match count looks low (for example, 5 of 50), check that you clicked the same column on both sides.

Stray spaces in the key would also lower the match count — but we trimmed our text in Workbook 2, so we should be safe.

Mission 4 — Merge with Couriers_Clean

Now we will bring courier_name and ship_class into our deliveries view.

1. Make sure Deliveries_Clean is still open in Power Query.
2. Click Home, then Merge Queries.
3. Click the courier_id column in the upper preview.
4. Choose Couriers_Clean from the lower drop-down.
5. Click the courier_id column in the lower preview.
6. Choose Left Outer as the Join Kind.
7. Confirm the hint at the bottom says '50 of 50 rows match'.
8. Click OK.

PAUSE AND CHECK

You should now have two new merged columns at the right: Routes_Clean and Couriers_Clean.

Both still hold a Table value in each row. We will expand them in the next mission.

Mission 5 — Expand the useful columns

Expanding turns a Table value into real columns we can read.

Expand Routes_Clean

1. Find the Routes_Clean column heading. To the right of the name there is a small Expand icon (two arrows).
2. Click the Expand icon.
3. Untick Use original column name as prefix.
4. Tick only route_name, sector, difficulty, base_distance_au.
5. Click OK.
6. Four new columns appear next to where Routes_Clean used to be.

Expand Couriers_Clean

7. Find the Couriers_Clean column heading and click its Expand icon.
8. Untick Use original column name as prefix if ticked.
9. Tick only courier_name, ship_class, member_tier.
10. Click OK.

PAUSE AND CHECK

Scroll along the columns. You should now see route_name, sector, difficulty, base_distance_au, courier_name, ship_class, member_tier — alongside the original delivery columns.

Look for the row with route_code R09. Its route_name and difficulty should be blank.

That blank is the unmatched row we expected — that is exactly how a Left Join is meant to behave.

Mission 6 — Try a small append

Appending stacks rows from another table. Imagine the Captain hands you a tiny extra batch of deliveries with the same columns.

We will not append a real second file in this pilot — we just want you to see where the option lives.

1. On the Home tab, click Append Queries.
2. Notice the dialog asks you to pick a table to stack on top of the current query.
3. Click Cancel for now. We are only exploring.

TRY THIS

Tiny challenge:

Look at the Append Queries icon on the ribbon. Find the small arrow next to it.

Click that arrow. Notice the option Append Queries as New.

Append as New keeps the original safe and makes a fresh combined query. Useful to remember.

Mission 7 — Validate the final result

Sanity-checking is the cherry on top.

1. In Power Query Editor, look at the bottom of the window for the row count. It should be 50.
2. Click View, then Column quality. Tick it on if it is off.
3. Find the route_name column. Hover the bar above it. The empty count should be 1.
4. Find the courier_name column. The empty count should be 0.
5. Click the status column. Tick the filter arrow to confirm only four values: Delivered, In Transit, Returned, Cancelled.

PAUSE AND CHECK

Row count: 50.

Blanks in route_name: 1 (this is the R09 record we expected).

Blanks in courier_name: 0.

If your numbers do not match, do not panic. Click each Applied Step from the top and watch what changes. The mismatch is somewhere there.

Mission 8 — Load the final table and visually check

We are going to load our final table to a sheet and add a small piece of conditional formatting.

1. In Power Query, change the query name to Deliveries_Final.
2. Click Home, then Close & Load and then select the whole table in the Excel sheet.
3. On the Home tab, click Conditional Formatting.
4. Click Highlight Cell Rules, then More Rules.
5. Choose 'Format only cells that contain' and set 'Format only cells with:' to Blanks
6. Choose a soft fill colour, click OK.
7. Blank cells will be highlighted gently — including the one in route_name for R09.

Now we are going to save this finished view as our final pilot output.

8. Click File, then Save As.
9. Save the file as 02_workbook_data/sd1_complete.xlsx.
10. Make sure the file type is Excel Workbook (*.xlsx).

NICE WORK

That is the end of Workbook 3.

You imported, cleaned, merged, appended (in spirit), validated and saved the final SD1 output.

If you want to compare, open 02_workbook_data/workbook3_complete.xlsx and 02_workbook_data/sd1_complete.xlsx.

What you can now do

- Explain merge and append in plain English.
- Choose the right Join Kind (usually Left Outer).
- Merge two queries on a shared key.
- Expand merged columns and pick which fields to keep.
- Read the row count and column quality to validate a join.
- Spot an unmatched row caused by a missing key.
- Load a final table to a worksheet and use conditional formatting.
- Save your final clean output to 02_workbook_data/sd1_complete.xlsx.

NICE WORK

Captain Vega has the answer to her question.

You have built a small, neat data pipeline from raw files to a clean view.

That is real Level 3 Data Technician work, made of small calm steps.

Be proud of yourself. The space waffles are on us.